McMaster University | Engineering

Robotic Intervention for Pollution & Plastic Litter Extraction (RIPPLE)

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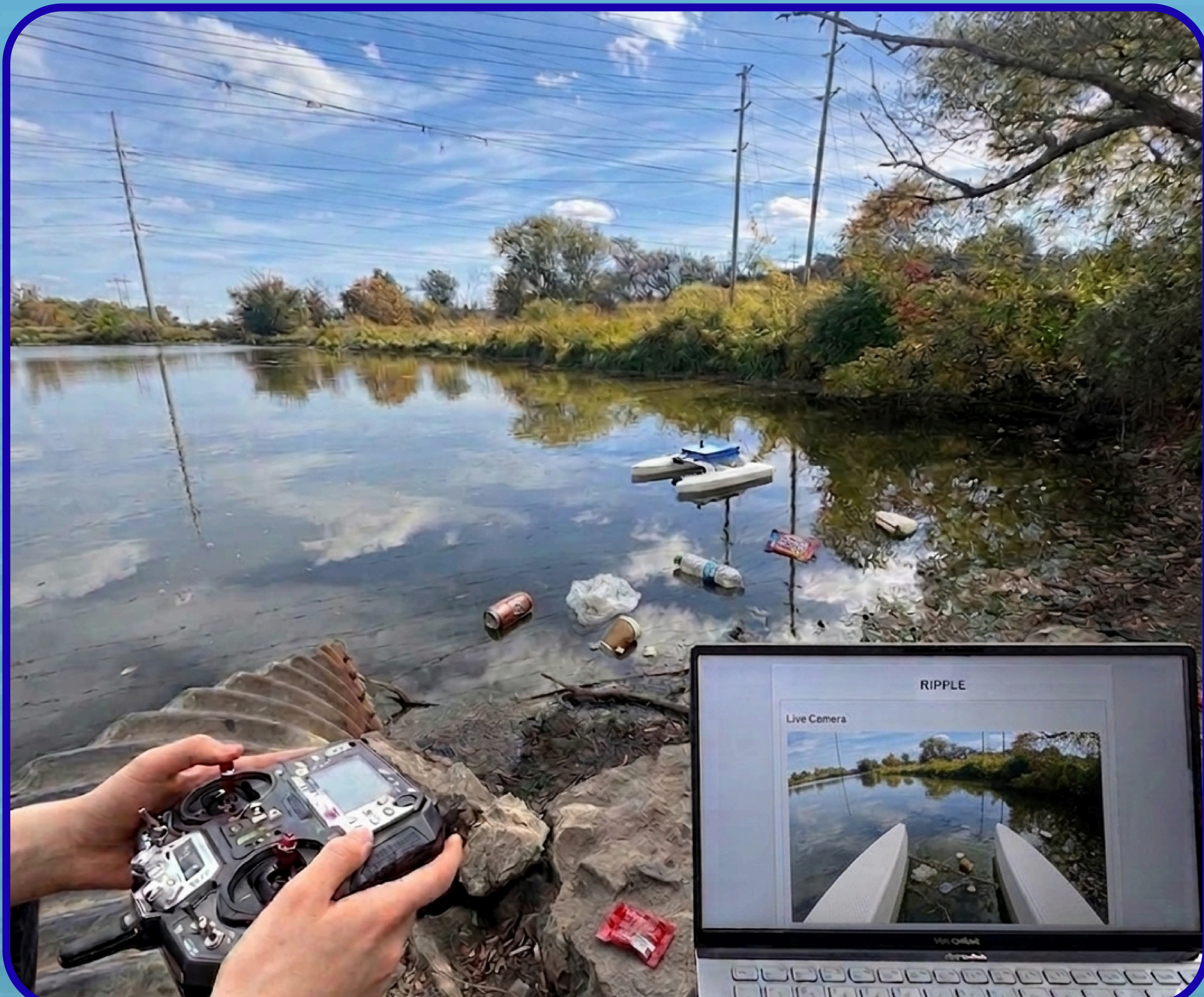
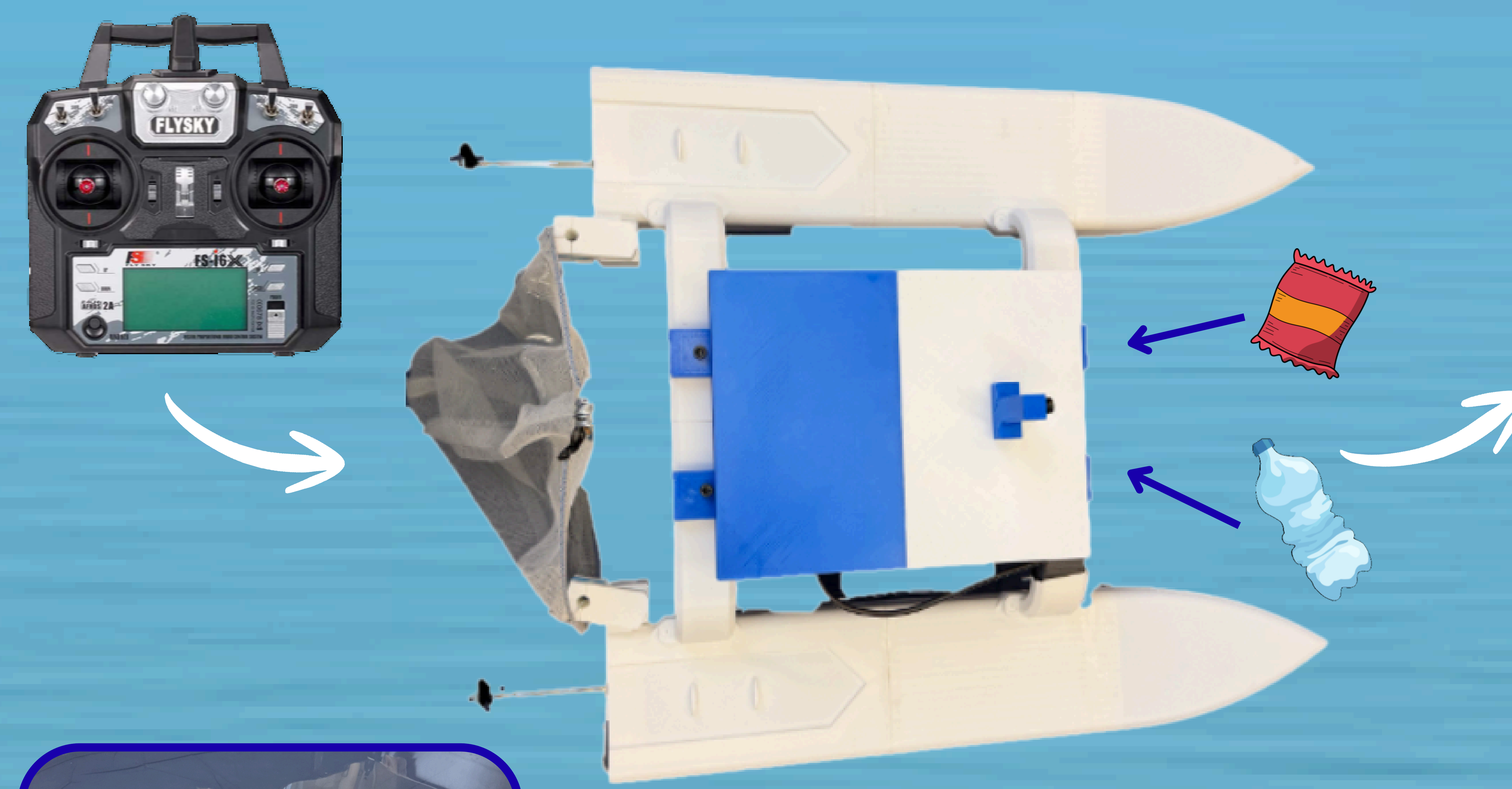
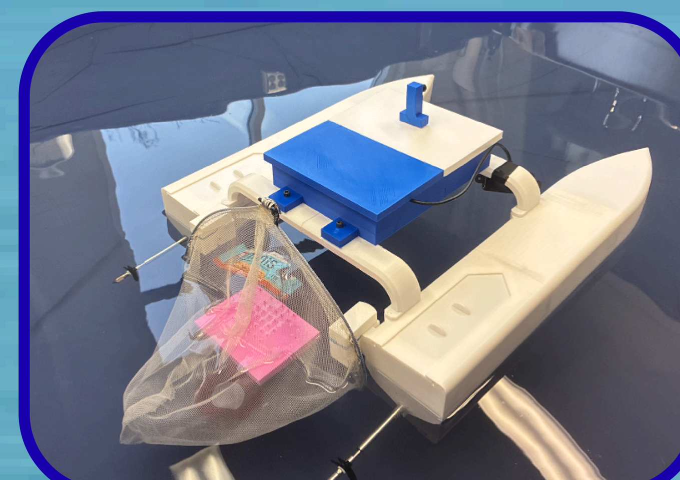






Motivation

- Surface water trash has been increasing in local ponds
- Damages wildlife, disturbs ecosystems, limits aquatic recreational activities & introduces microplastics into our freshwater
- **Objective:** Collect **small, surface water trash** in local ponds

What Is RIPPLE?

- **Remotely controlled** robotic system
- Designed to **collect small floating trash** on the surface of local ponds

Key Features

- Remote control surface navigation
- Passive debris collection system
- Real-time **Camera + UI** feedback
- **GPS-based** positioning

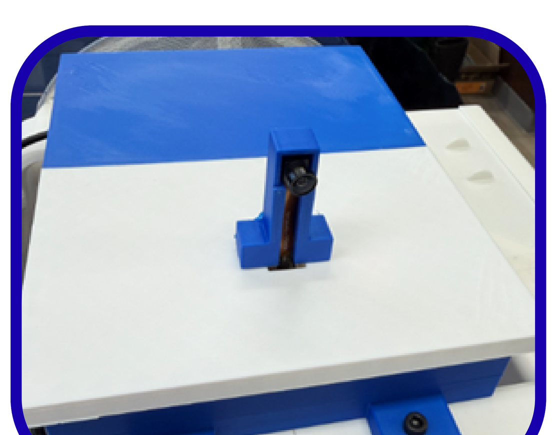
Motor Control

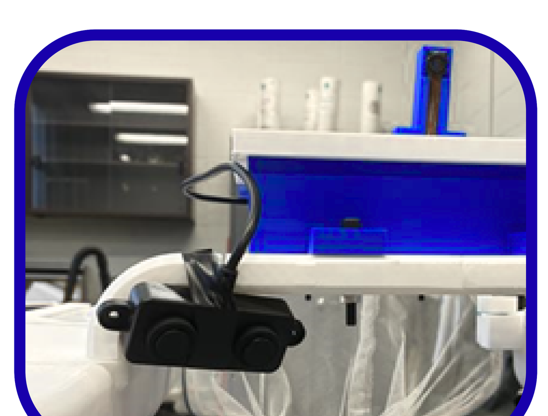
- User is provided a **remote control**
- **Joystick** controls boat's direction
- **Auxiliary switch** controls boat's speed

MCU: Camera, GPS & Proximity Sensor

System has three input peripherals connected to an **MCU** which sends data to the **UI**:

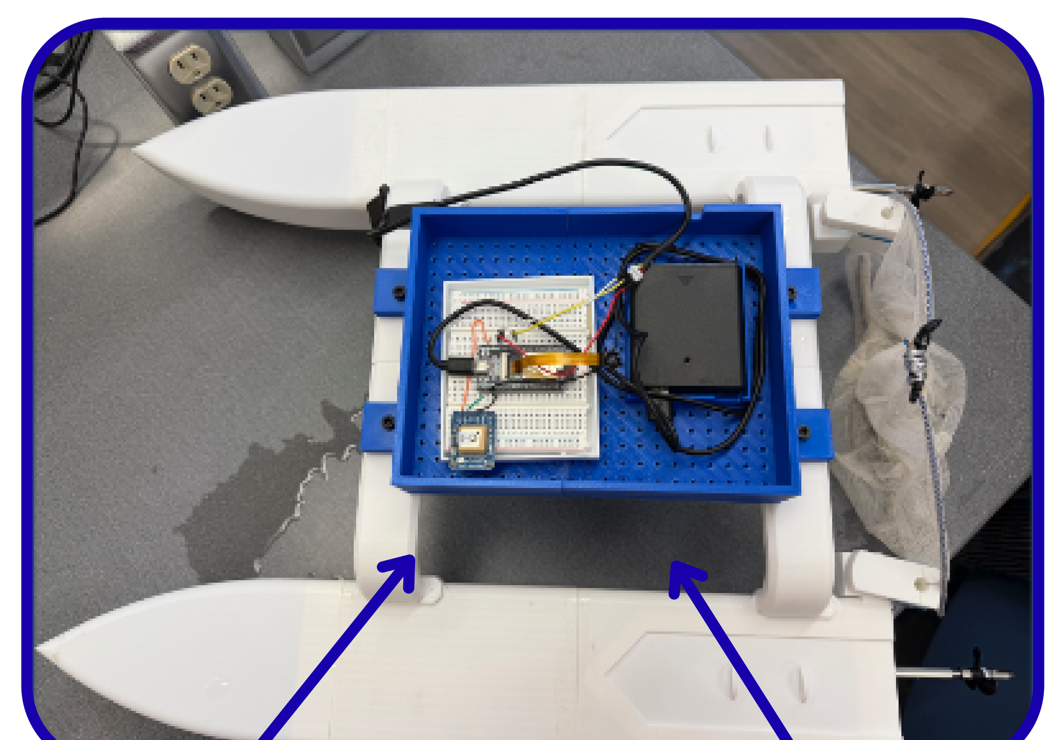
- **Camera** to show RIPPLE's field-of-view
- **GPS** to track its location
- **Proximity Sensor** to make sure trash flows into the net

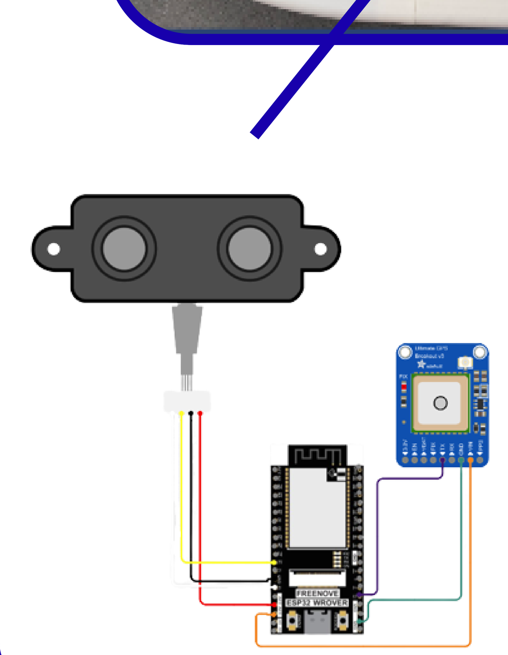


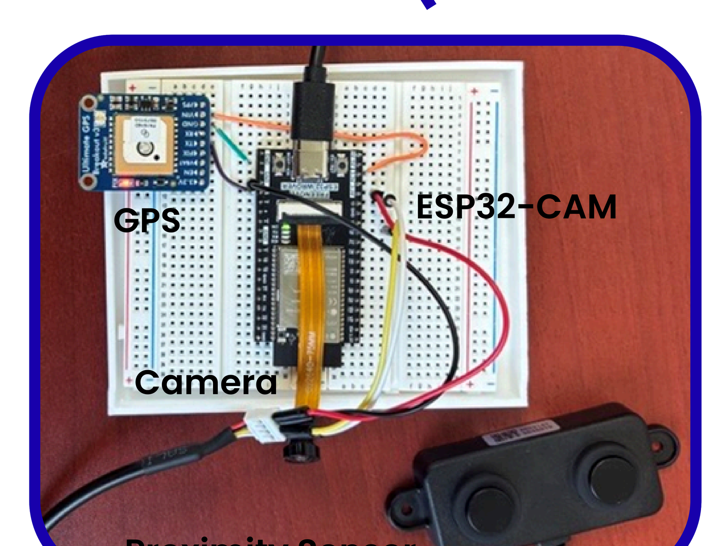


Boat Hull

- **Twin-hull catamaran design** for high stability
- **Pointed bow** to reduce drag
- Central plate & side compartments host the MCU & **motors**
- **Forward-facing passive net** system that captures floating surface trash

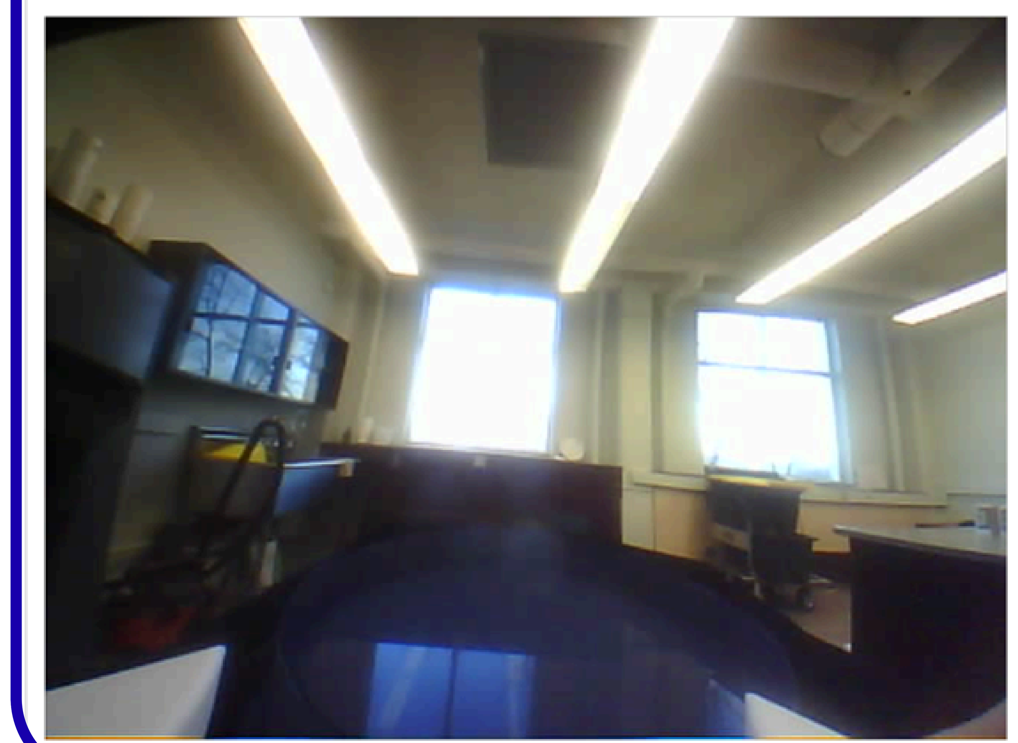






UI Visualization

Live Camera



Proximity Sensor

Object Distance: 1113 mm

Status: Clear

Proximity Sensor

Object Distance: 329 mm

Status: OBJECT DETECTED

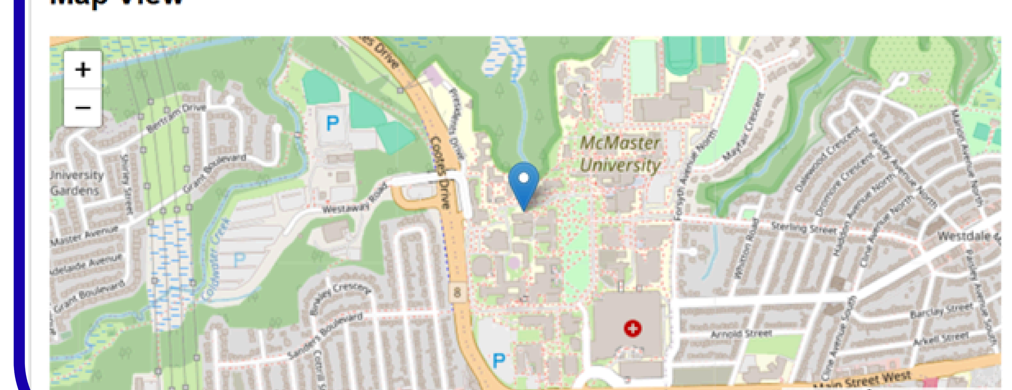
GPS Location

Latitude: 43.262470

Longitude: -79.921150

Satellites: 3

Map View



User Interface (UI)

- Accessed on any device connected to **WiFi** via a browser
- **Livestream** of the **Camera**
- **Proximity Sensor** distance
    - **Warning** when object is detected within **0.5 m**
- **Latitude / longitude coordinates**
- Moving marker on a **live map**